

- Laboratory Assignment - *Digital Signatures and Digital Certificates*

Disclaimer: During this course/lab you will be introduced to various notions of computer security, with the aim of learning how to secure systems. All notions and exercises are presented for didactic purposes, even if sometimes you are supposed to think like an adversary. Do not use these techniques for malicious purposes! They may have legal consequences in the event of committing crimes, for which you become fully responsible!

1. Basic Concepts



Recall *digital signatures* and *digital certificates* [1,2].



The following questions refer to the digital certificate of the faculty website [3]:

- a) Who issued the digital certificate? Let's Encrypt/ R13
- b) What is the validity of the certificate? 25 Feb 2026 00:38:20 GMT - 26 May 2026 00:38:19 GMT
- c) How many bits is the public key defined in? 2048
- d) What is the value of the encryption exponents in the certificate and in the certificates that attest it in the chain? What do you notice? Does this impact security?
65537, 65537
It is standard, it shouldn't impact security

2. Key Generation using Putty



Use *Putty* [4] to generate SSH keys:

- a) Download *puttygen.exe* [4].
- b) Generate a 2048-bit RSA public key - private key pair. Press *Generate*.
- c) Add *PassPhrase*. What is this for?
- d) Export the public key to the *public_key.pub* file, the private key to the *private_key.ppk* file. To do this, use *Save public key*, respectively *Save private key*.
- e) Export the key in *openssh* format. To do this, go to *Conversions* and *Export OpenSSH key*.
- f) Open and see what all the generated files contain.

3. Digital Certificate Generation using OpenSSL



Answer the following questions using *OpenSSL* [5]:

- a) Generate an RSA key.
- b) Use the previously generated key in a *self-signed* certificate, valid for 120 days, stored as *ca.crt*. Use the following information:

Country: RO

Province: Muntenia

City: Bucuresti

Organization: CA_SSI

Department: CA_SSI_Lab

Common Name: CA_numele vostru (e.g.: CA_Andrei)

E-mail: test@test.ro

For this, use the next command:

```
openssl req -new -x509 -days <days> -key <key> -out ca.crt
```

- a) Read about *X.509* [6]. Have a look at the digital certificated that was created:

```
openssl x509 -text -noout -in ca.crt
```

- b) Use this CA certificate to sign/issue another certificate of a subordinate entity SUB_SLA. Use the following information:

Country: RO

Province: Muntenia

City: Bucuresti

Organization: SUB_SSI

Department: SUB_SSI_Lab

Common Name: SUB_numele vostru (e.g.: CA_Andrei)

E-mail: test_sub@test.ro

To do this, first generate a new 2048-bit SUB_SSI entity key in the *sub.key* file.

- c) Initiate a *Certificate Signing Request (CSR)* *sub.csr*:

```
openssl req -new -key sub.key -out sub.csr
```

- d) Then create a certificate for SUB_SSI *sub.crt* signed by the CA authority, valid for 60 days, with serial number 02:

```
openssl x509 -req -days <days> -in sub.csr -CA <certificat_CA> -  
CAkey <ca_key> -set_serial <serial_no> -out sub.crt
```

- e) View the created digital certificate:

```
openssl x509 -text -noout -in sub.crt
```

- f) Transform this certificate to *PKCS#12*:

```
openssl pkcs12 -export -out sub.p12 -inkey sub.key -in sub.crt -  
chain -CAfile ca.crt
```

- g) Verify the content of *sub.p12* by using:

```
openssl pkcs12 -info -in sub.p12
```

4. Digital Certificates in the Post-Quantum Era



Read about the impact of quantum computers on the security of digital certificates. Some easy-to-read resources are available at [7-11].



Answer the following questions:

- a) How can you explain the quantum threat? What is the impact on the security provided by current digital certificates?
- b) What measures/solutions can decrease risks and adopt post-quantum digital certificates?
- c) What challenges can you identify in adopting post-quantum digital certificates?

References

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2. Itfreetraining. *What are certificates?* Available at: https://www.youtube.com/watch?v=LRMBZhdfjDI&ab_channel=itfreetraining Last accessed: February 2026
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7. NIST. Post-Quantum Cryptography. Available at: <https://csrc.nist.gov/projects/post-quantum-cryptography> Last accessed: February 2026
8. Webnic. Beyond SSL: How Digital Certificates Are Evolving in the Post-Quantum Era. Available at: <https://webnic.cc/cyber-security/beyond-ssl-how-digital-certificates-are-evolving-in-the-post-quantum-era/> Last accessed: February 2026
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10. Thales. Preparing Digital Identity for the Post-Quantum Era. Available at: <https://www.thalesgroup.com/en/news-centre/insights/public-security/civil-identity/preparing-digital-identity-post-quantum-era> Last accessed: February 2026
11. Tencent Cloud. What are the quantum-resistant digital signature standards in terminal security? Available at: <https://www.tencentcloud.com/techpedia/123051> Last accessed: February 2026

- a) Quantum computers can break the mathematical foundations of current digital certificates. They use algorithms like Shor's to quickly solve problems that traditional computers can't, making RSA and ECC encryption vulnerable. This means quantum computers could forge certificates, decrypt communications, and compromise online security.
- b) Implementing post-quantum cryptographic algorithms resistant to quantum attacks, using hybrid certificates (combining traditional and quantum-safe methods), increasing key sizes temporarily, developing crypto-agile systems, and following NIST's post-quantum cryptography standards.
- c) Challenges include larger key sizes requiring more storage and processing power, compatibility issues with existing systems, lack of mature standards, performance impacts, high migration costs, uncertainty about which algorithms will become standard, and the need for new hardware to support these certificates.